BACKGROUND & METHODOLOGY

Adaptive Geodesic Conformal Prediction for Spherical Harmonic Climate Reconstruction

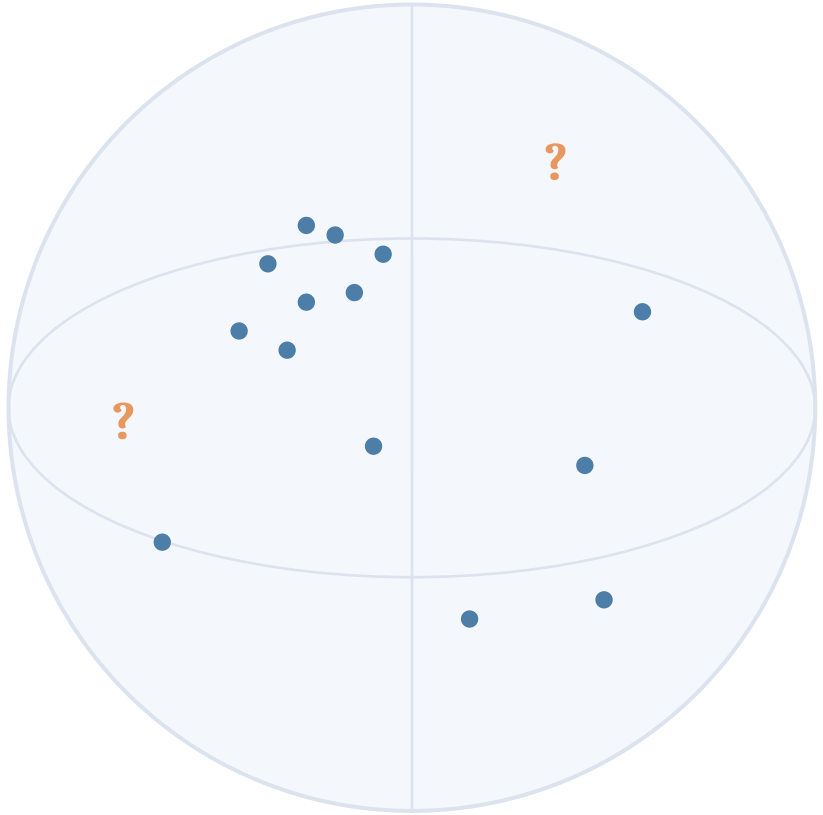
Uncertainty: Reconstruct Temperature From Scattered Observations to Trustworthy Climate Maps

Why the problem matters, then explain methodology -> contribution -> result

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Reconstructing continuous temperature field from scattered dots

- Many things we care about — temperature, pressure, magnetic fields — vary smoothly across the whole surface of the Earth.
- But **we only observe** them at **a limited number of locations**: weather stations, cities, sensors.
- The task is to fill in the gaps: reconstruct a continuous field over the entire sphere from these sparse, **unevenly distributed** observations.
- **Coverage is not uniform** — dense in places like Europe, very sparse over oceans, mountains, and remote regions.
- **The challenge** is **to understand where those predictions are reliable**.



Blue dots = observation cities · gaps = unobserved regions

WHY THIS MATTERS

A Prediction Without Trust Isn't Enough

1

Decision Making in applications

Reconstructed fields feed into agriculture, environmental monitoring, and other downstream models — decisions are only as good as the map they're based on.

2

Accuracy isn't uniform

The model can be excellent where data is in densely observed regions , and far less reliable where it is sparse — e.g. oceans, deserts, high mountains.

3

RMSE hides this variation

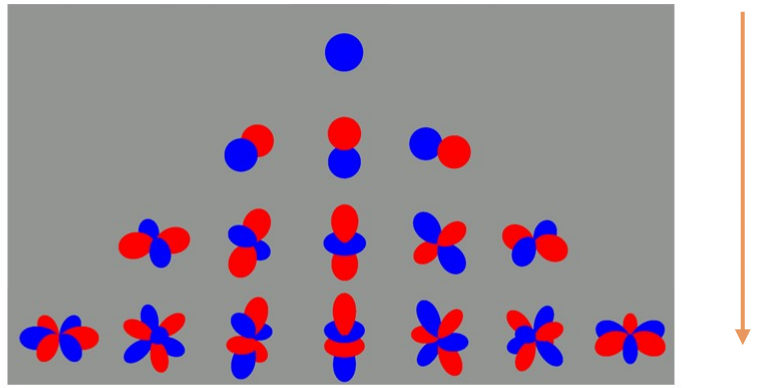
Being told the model is ‘90% accurate on average’ doesn’t tell you whether to trust the prediction at any one specific location.

This dissertation focuses on giving every prediction a local, spatially-aware measure of how much to trust it — not just an overall average. That's the gap I address with spatially adaptive uncertainty quantification.

Spherical Harmonics — Reconstructing the Field

- Building on prior work (Thomas, 2025), the field is **represented as a sum of smooth, wave-like patterns wrapped around the sphere.**
- **The spherical harmonic degree L**, determines how many fixed basis functions are included. Fewer basis functions give a smoother, large-scale reconstruction, while more basis functions allow finer spatial detail.
- **The regularisation parameter λ,** controls the strength of the penalty on overly complex reconstructions, helping to prevent overfitting.
- This dissertation uses this existing model unchanged as its starting point — the base predictor.

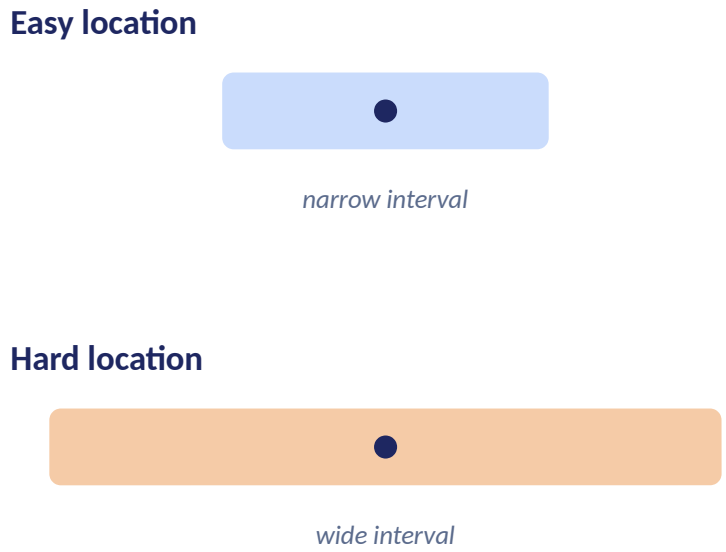
increasing degree, L — model complexity



Example figure from (Thomas, 2025) Figure 2: Spherical harmonics for degrees 0 to 3. These are examples of the fixed basis functions used to represent the temperature field.

Conformal Prediction — Knowing How Much to Trust Each Point

- Conformal prediction wraps around any existing predictor and adds a **calibrated interval** around each prediction.
- It comes **with a guaranteed coverage rate** (e.g. 90% of the time the true value falls inside the interval) — without needing to know how the underlying model works internally.
- Unlike Gaussian Processes, it makes no assumption about the shape of the model's errors — it only needs a held-out calibration set.
- This dissertation's key idea: **let the interval widen or narrow depending on how difficult each location is** — instead of one fixed size everywhere.



Same predictor, same target coverage — interval width adapts to local difficulty.

OUR CONTRIBUTION

Adapting Geodesic Conformal Prediction to Scalar Fields

Prior work

Amiri Shahbazi & Baheri (2026)

Designed for manifold-valued responses — e.g. a direction on a sphere**. Geometry is used to measure error on the OUTPUT.**

→

This dissertation

Scalar climate reconstruction

The **response (temperature) is scalar**, not manifold-valued. **Geometry is instead used on the INPUT side** — to define which locations are 'nearby' when estimating local difficulty.

In short, this dissertation contributes:

- A **methodological adaptation, new local difficulty estimator $\hat{\sigma}(x)$, empirical validation, robustness checks** across model complexity, and a **geographic interpretation** of where uncertainty concentrates.
- Used geographical distance on the sphere to identify nearby locations and estimate local prediction difficulty, sigma-hat(x).
- Used this difficulty estimate to adapt the conformal interval at each location.

adapt conformal prediction to spherical temperature reconstruction by using geodesic neighbourhoods to estimate local prediction difficulty

RESULTS

Does this adaptive approach actually improve uncertainty estimates?

Results · Sze Ying Choong · MSc Data Science

BASELINE COMPARISON

Marginal Coverage — Looks Similar...

Method	Marginal Coverage	Mean Interval Width
Standard (non-adaptive)	90.9%	4.65 °C
Adaptive geodesic (proposed)	93.7%	4.59 °C

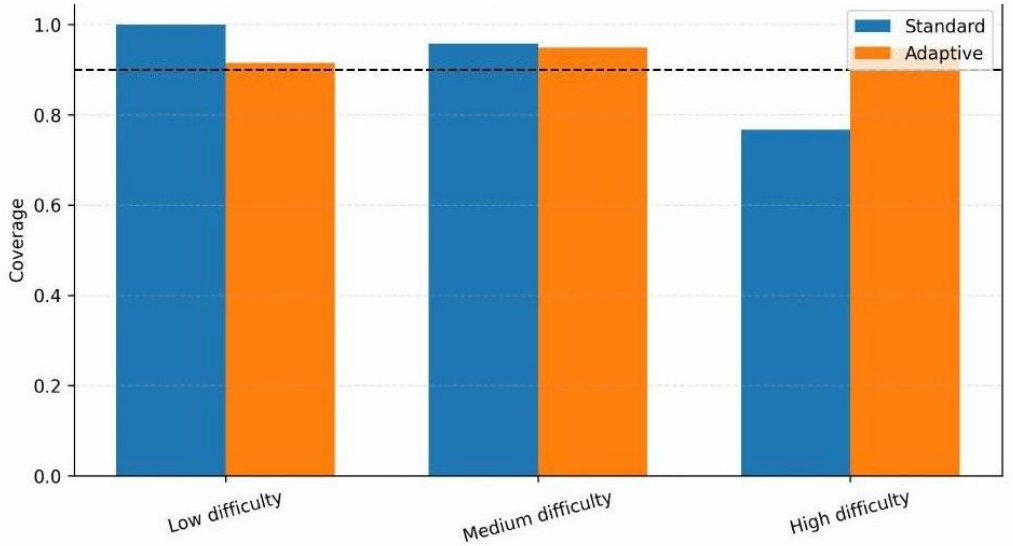
- Both methods land close to the 90% nominal target — and interval widths are nearly identical
- **So marginal coverage alone makes the two methods look basically equivalent...**
- **...that's exactly the problem. It hides what's actually happening underneath.**

MAIN RESULT

Coverage by Difficulty Tertile

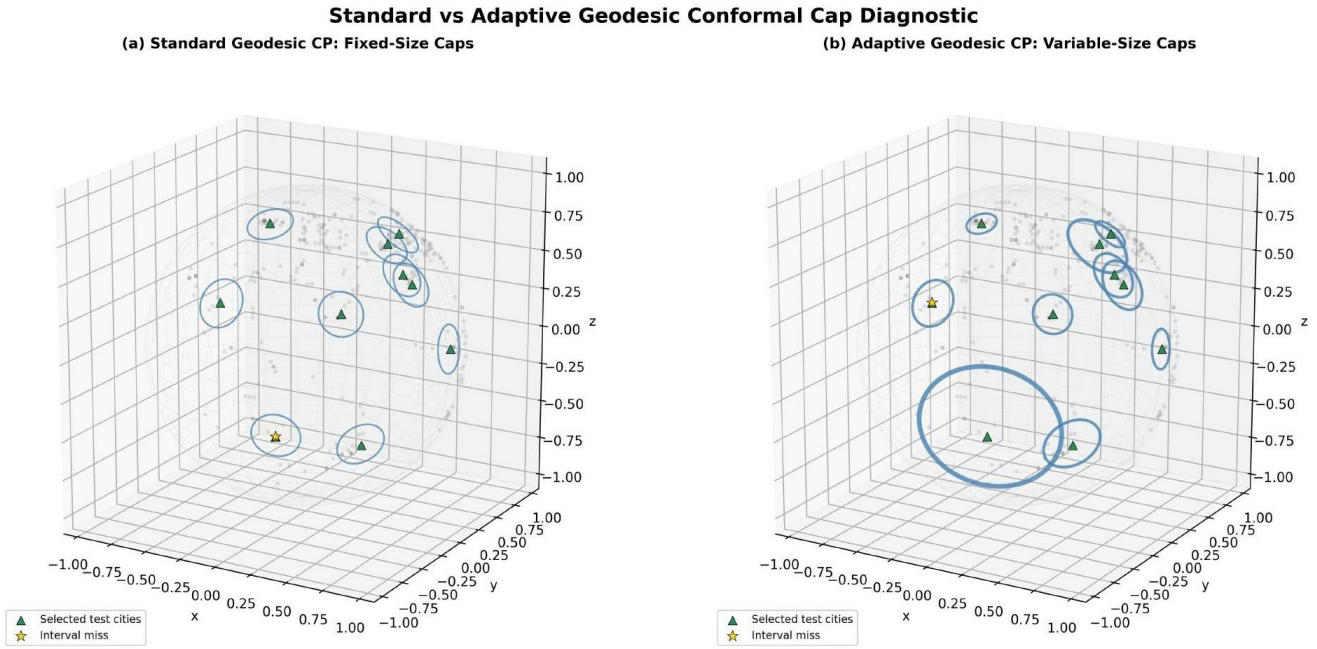
Tertile	Standard	Adaptive	Adapt. Width
Low difficulty	100.0%	91.5%	1.48 °C
Medium difficulty	95.7%	94.9%	3.89 °C
High difficulty	76.7%	94.8%	8.46 °C

- Standard over-covers the easy tertile (100%) and badly fails the hard tertile — 76.7%, a 13-point miss below target
- **Adaptive stays within 91–95% across all three — right where reliability matters most**
- Not from inflating width overall — it's reallocated: 1.48°C on easy locations, 8.46°C on hard ones



Coverage by difficulty group vs 90% target (dashed line)

Same Result, Seen on the Sphere



- Left: standard method — every cap the same fixed size
- **Right: adaptive method — caps grow in high-difficulty areas, shrink in easy ones**
- Gold stars = coverage misses — adaptive doesn't eliminate misses, it reallocates where the risk sits

VALIDATION

Is $\hat{\sigma}(x)$ Actually Meaningful?

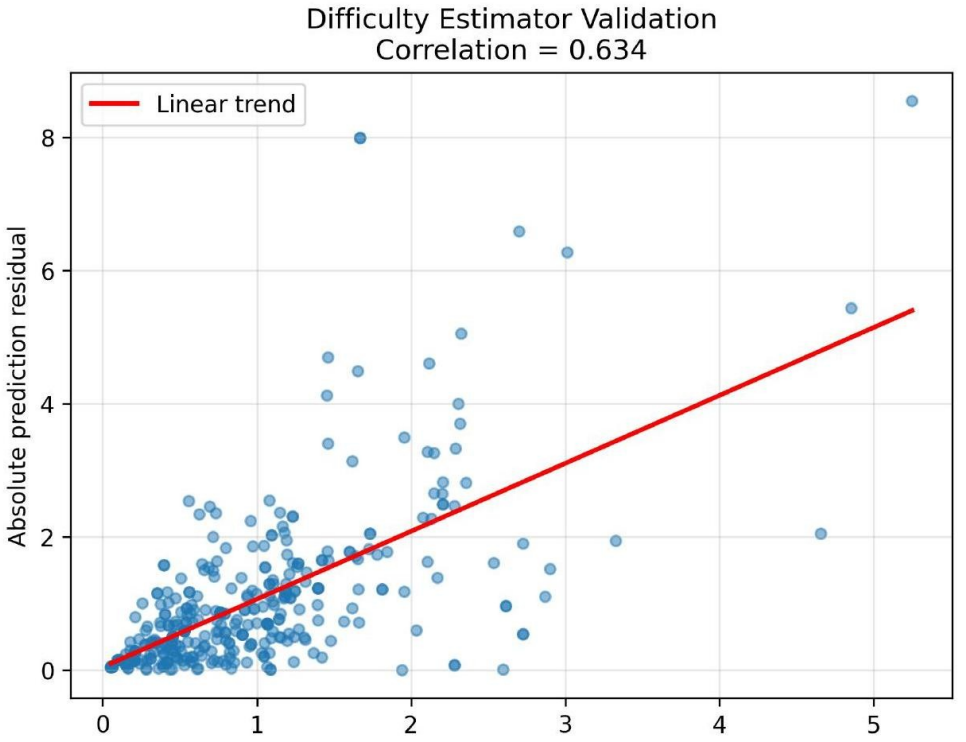
- Pearson correlation between $\hat{\sigma}(x)$ and true absolute residual:

$r = 0.634$

- Higher than the source paper's real-world case study (r = 0.516)
- **Two plausible explanations — not mutually exclusive:**

1. **Easier setting**: exact coordinates, scalar response
2. Temperature is more **spatially autocorrelated** than geomagnetic anomalies

- Can't fully separate these two with the current data — flagged directly in the Discussion



STRESS TEST

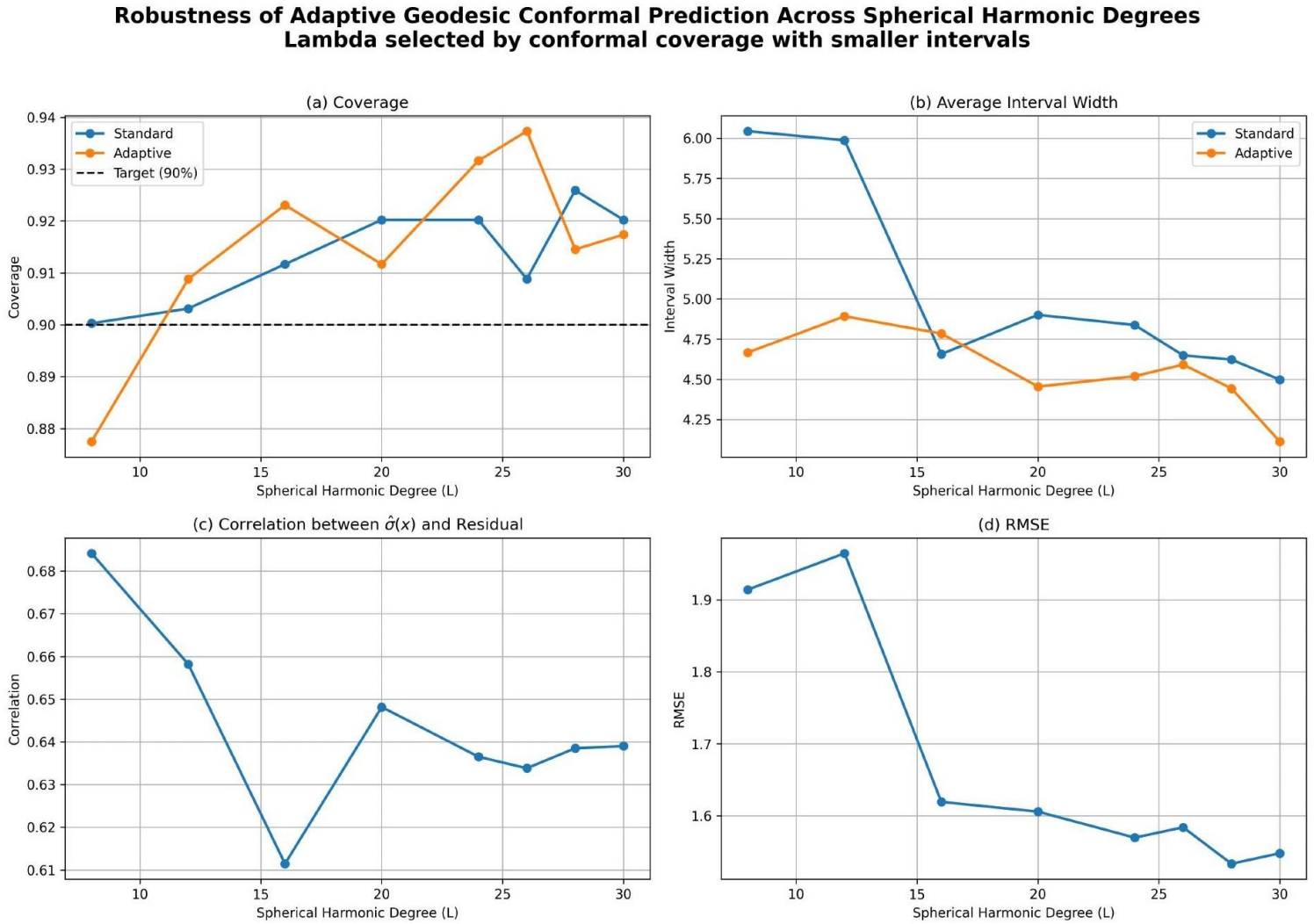
Six Difficulty Bins Instead of Three on Adaptive Method

Bin	n	Mean $\hat{\sigma}$	Coverage
1 (easiest)	59	0.185	94.9%
2	59	0.435	88.1%
3	59	0.663	89.8%
4	58	0.969	100.0%
5	57	1.256	96.5%
6 (hardest)	59	2.267	93.2%

$\hat{\sigma}$ = estimated local difficulty (defined earlier) — higher means the model expects a harder prediction

- Coverage ranges from 88.1% to 100% across six bins
- **Worst bin: 88.1% (only 1.9% points below the 90% target)**
- Compare to the standard method's worst tertile: 76.7%
- **Pattern holds at finer resolution** — not an artefact of picking three groups

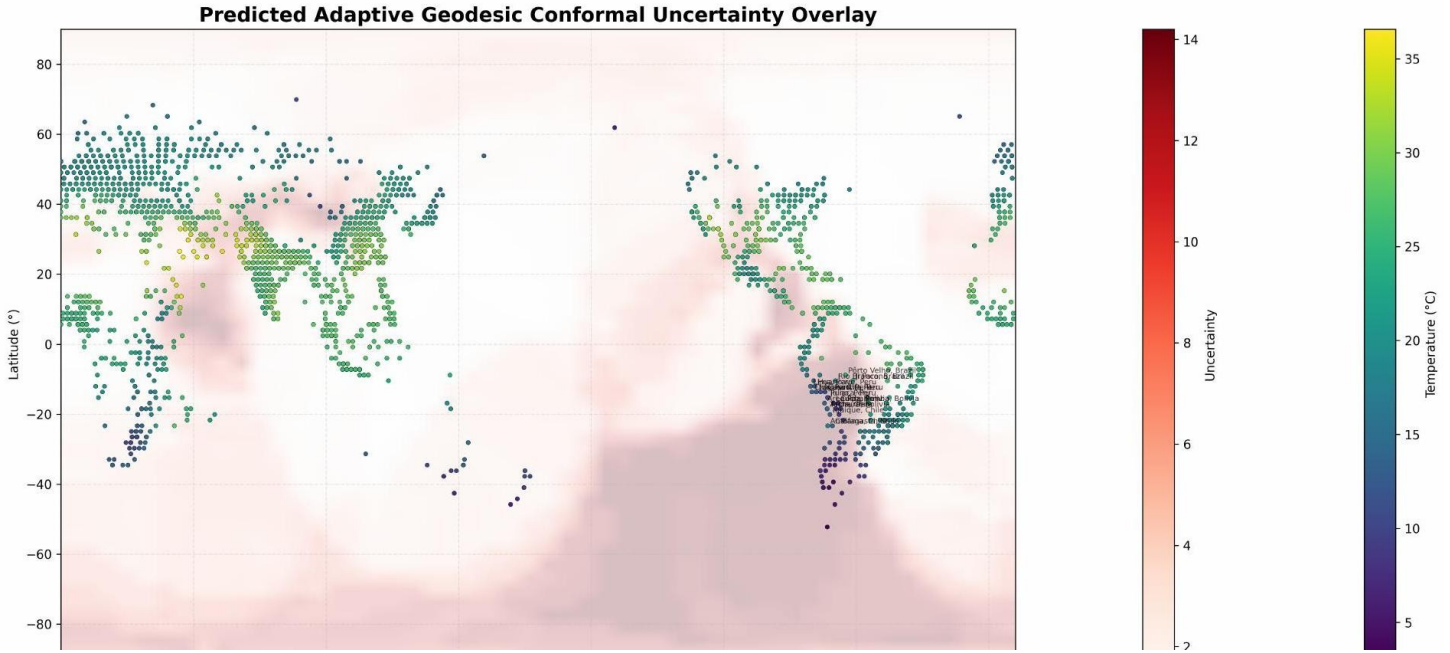
Does This Hold Across Model Complexity?



- Repeated the full pipeline at L = 8 through L = 30
- **Adaptive coverage stays close to 90% across all degrees**
- Correlation r stays in a tight band — 0.61 to 0.68
- Two different λ-selection criteria give consistent results
- **Not a fluke of choosing L=26 specifically**

GEOGRAPHIC INTERPRETATION

Where Is Uncertainty Concentrated?



- 17 of the 20 highest-uncertainty cities are in the high-altitude Andes (Bolivia, Peru, N. Chile)
- **These cities typically have only 2–5 neighbours within 250 km — much sparser than Europe or E. China**
- Plausible story: sparse observations + high-elevation climate variability the model struggles to capture — but correlation, not proven causation

WHAT THIS MEANS

Three takeaways

1

Standard conformal fails where it matters

Badly under-covers hard regions (76.7%). Adaptive fixes this by reallocating width, not inflating it.

2

$\hat{\sigma}(x)$ is informative

r = 0.634, and the pattern is geographically interpretable — concentrated in the high Andes.

3

The result remains robust

Holds across spherical harmonic degrees L = 8 to L = 30, with two different selection criteria.

a spatially adaptive uncertainty layer that gives more balanced uncertainty estimates across the temperature field.

THE END……

Thank You !

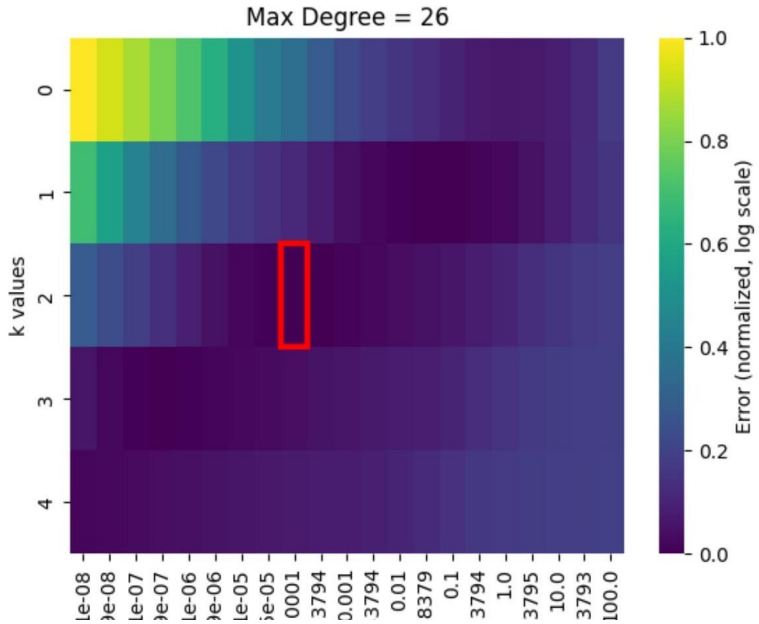
Move on to Q & A session……

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SETUP

Hyperparameter Selection

- 5-fold cross-validated grid search over λ and Sobolev order k, at fixed degree L = 26
- **Minimum error at $\lambda = 10^{-4}$, k = 2**
- Low-error region is broad and shallow — not a fragile choice
- This configuration is used throughout every result that follows



From Raw Observations to Adaptive Intervals

Berkeley Earth Temperature Data
3,510 cities, July 2003

↓

Preprocessing & Data Splitting
Train / Validation / Calibration / Test

↓

Spherical Harmonic Regression
Reconstruct the temperature field

↓

5-Fold Cross-Validation
Generate out-of-fold residuals

↓

Geodesic k-NN Difficulty Estimation
$\hat{\sigma}(x)$, $k = 20$

↓

Adaptive Conformal Calibration
Normalise calibration residuals by $\hat{\sigma}(x)$

↓

Adaptive Prediction Intervals
Wider where difficulty is high, narrower where low

Standard CP	Adaptive CP
One global threshold, $\hat{q}$	Local difficulty, $\hat{\sigma}(x)$
Same interval width everywhere	Variable interval width
No spatial adaptation	Geodesic spatial adaptation

What the Results Mean — and What They Don't

- Standard conformal's **failure isn't just a number** — it maps onto a real spatial pattern, concentrated in sparse, high-difficulty regions like the Andes.
- The **adaptive method's advantage doesn't come from wider intervals overall** — it specifically reallocates width to where it's needed, which is why the average width barely changes.
- Conformal prediction was chosen over a Gaussian Process because it **needs no covariance/kernel assumption** — the coverage guarantee holds even if the base spherical harmonic model isn't perfectly specified.
- **Uncertainty maps only reflect the observed data regime** — calibration doesn't extend to genuinely unobserved regions, like the open ocean, where there's no nearby data to calibrate against.

Where This Analysis Has Known Gaps

- The underlying Berkeley Earth data is a processed, spatially-interpolated estimate, **not a raw station measurement** — residuals reflect fit to that processed field**, not to physical truth.**
- $\hat{\sigma}(x)$ is estimated from only 2,106 training points using ordinary random 5-fold CV, **not spatially-blocked folds — some risk of optimistic bias** from spatial autocorrelation between train and validation points.
- The neighbourhood size, $k = 20$, **was inherited directly** from the source paper rather than independently tuned on this dataset.
- The coverage guarantee **follows standard Euclidean split-conformal theory** — this is a structural adaptation of the geometric idea, not a re-proof of the original manifold-response guarantee.
- All results come from a single fixed calibration/test split ($n = 351$) — **no repeated-trial variance estimates**, so numbers carry unquantified sampling uncertainty.